

Lec 13-2, Wed, Dec 3

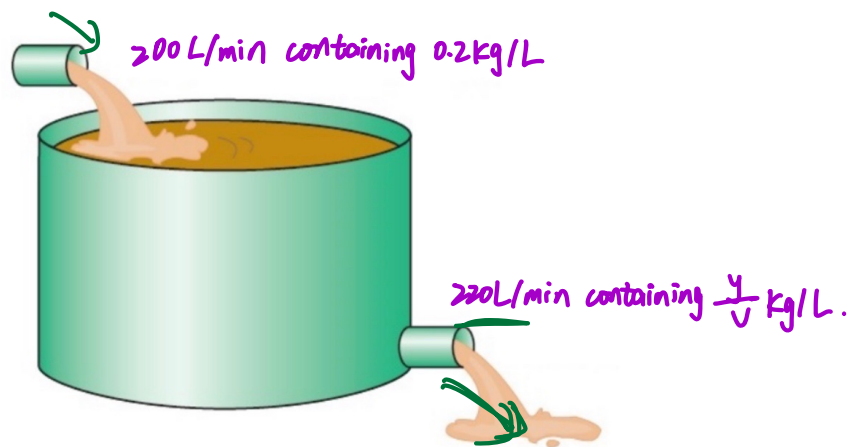
- Applications of D.E. (9.3)
 - Mixture problems
 - Orthogonal trajectories
- Autonomous equations and phase line analysis (9.4, 9.5)
 - stable and unstable equilibria
- Euler's method (9.1)
- Review of Calculus I.

Mixture problems

Consider a container satisfying the following conditions:

- It initially contains 10000 L of solution, having 50 kg of salts dissolved in it.
- The solution leaks out of the contain at a rate 220 L/min.
- A solution with salt, whose concentration is 0.2 kg/L, is pumped into the container at a rate of 200L/min.

Suppose that the newly added solution is **instantly** well mixed with the solution that was already in the container. **What is the amount of salt in the container 20 minutes after the initial time?**



$$\begin{aligned} \bullet \quad V(0) &= 10000 \text{ (L)} & \Rightarrow V(t) &= 10000 - 20t \text{ (L)} \\ V'(t) &= -220 + 200 = -20 \text{ (L/min)} \end{aligned}$$

• The weight of salt is $y(t)$.

$$\text{— } y(0) = 50 \text{ (kg)}$$

$$\text{— } y'(t) = -220 \cdot \frac{y(t)}{V(t)} + 200 \cdot 0.2 \quad (\text{kg/min})$$

$(\text{L/min}) \cdot (\text{kg/L}) \quad (\text{L/min}) \cdot (\text{kg/L})$

$\Rightarrow$

$$y'(t) = 40 - 220 \frac{y(t)}{V(t)} = 40 - 220 \frac{y(t)}{10000 - 20t}$$

$\Rightarrow$

$$\left\{ \begin{array}{l} y(0) = 50 \text{ (kg)} \\ y'(t) = \frac{-220}{10000 - 20t} y(t) + 40. \end{array} \right.$$

$$\Rightarrow y(t) = \dots \Rightarrow y(20) = \dots$$

□.

• Orthogonal trajectories

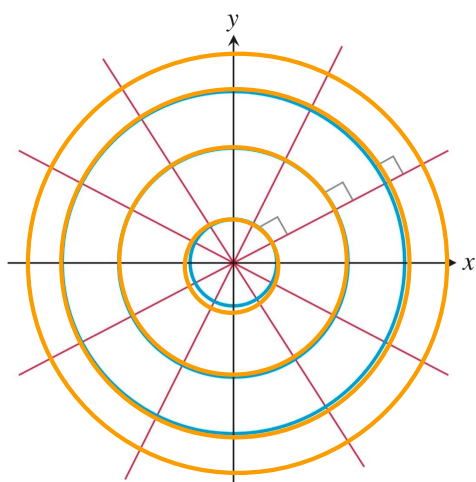
Let S be a family of curves. A curve C is called an orthogonal trajectory of S if C intersects every curve in S orthogonally (i.e. $\perp$).

e.g. Let $S \triangleq \{x^2 + y^2 = a : a > 0\}$ be the family of circles centered

at $(0,0)$. Then any line $y = kx$ is

an orthogonal trajectory of S .

Hence, $\{y = kx, k \in \mathbb{R}\}$ is a family of orthogonal trajectories of S .

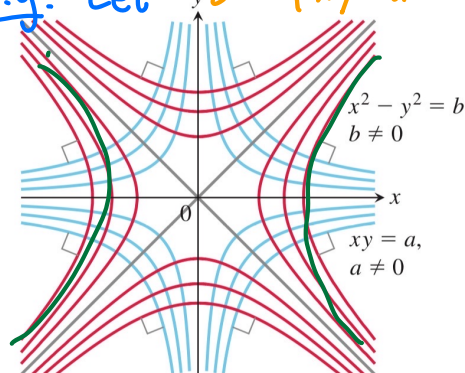


To find a family of orthogonal trajectories of a given family S :

Step 1 : find a D.E. that all curves in S satisfy ;

Step 2 : use the fact that if $L_1 \perp L_2$ and L_1 has slope m , then L_2 has slope $-\frac{1}{m}$.

e.g. Let $S \triangleq \{xy = a : a \neq 0\}$. Step 1: Find a D.E.!



$$xy = a \Rightarrow 1 \cdot y + x y' = 0 \Rightarrow y' = -\frac{y}{x}.$$

Step 2: A curve is an orthogonal trajectory of S

if and only if it satisfies $y' = \frac{-1}{-\frac{y}{x}} = \frac{x}{y}$

for all (x, y) with $x \neq 0$ and $y \neq 0$.

Step 3: solve new D.E: $y' = \frac{x}{y}$.

$$\Rightarrow \frac{dy}{dx} = \frac{x}{y} \Rightarrow y dy = x dx$$

$$\Rightarrow \int y dy = \int x dx$$

$$\Rightarrow \frac{1}{2} y^2 = \frac{1}{2} x^2 + C_1$$

$$\Rightarrow y^2 = x^2 - C \quad (\text{where } C = -2C_1 \text{ is any real number})$$

$\therefore \{x^2 - y^2 = C : C \in \mathbb{R}\}$ is a family of orthogonal trajectories of S .

[hyperbola if $C \neq 0$;

a pair of lines if $C = 0$] $\Rightarrow x^2 - y^2 = 0 \Rightarrow y = \pm x$.

• Autonomous equations and Phase line analysis

Consider the D.E. $y' = (y+1)(y-2)$.

Def. An autonomous equation is an D.E. of the form

$$\frac{dy}{dx} = f(y)$$

e.g. They are all autonomous :

$$\cdot \frac{dP}{dt} = k \cdot P \quad [\text{Malthusian growth model}]$$

$$\cdot \frac{dP}{dt} = k \cdot P \left(1 - \frac{P}{M}\right) \quad [\text{Logistic growth model}]$$

$$\cdot \frac{dH}{dt} = k \cdot (H - R) \quad [\text{Newton's law of cooling}]$$

· If we think of $y = y(t)$ as the position of a moving particle on the y -axis at time t , then $\frac{dy}{dt} = f(y)$ is stating that the velocity of the particle depends only on the position y , but not the time t .

Q: How do the solutions to $y' = f(y)$ behave?

Suppose k is a root of f , i.e., $f(k) = 0$.

Consider the constant function $y \equiv k$.

· $\frac{dy}{dx} = 0$, $\forall x$.

· $f(y(x)) = f(k) = 0$, $\forall x$.

Hence, the constant function $y \equiv k$ is a solution to $y' = f(y)$.

Def: Given an autonomous equation $\frac{dy}{dx} = f(y)$, for any root k of f

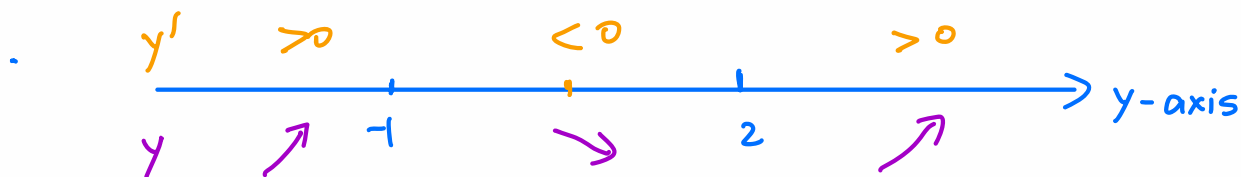
· k is called an equilibrium value;

· The constant function $y \equiv k$ is called an equilibrium solution to the autonomous equation.

We can analyze the behavior of the solutions by drawing all equilibrium values on the y -axis and analyze the signs of derivatives on the intervals separated by the equilibrium values.

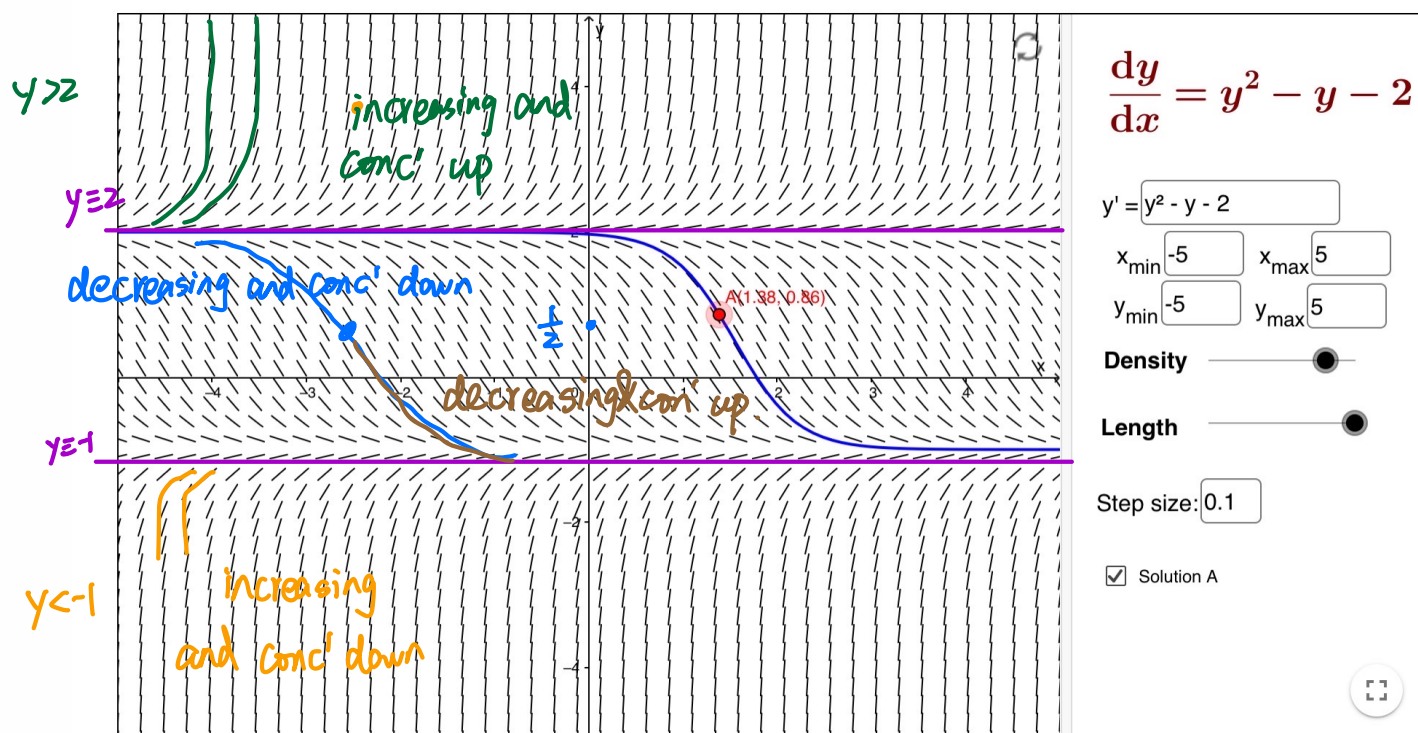
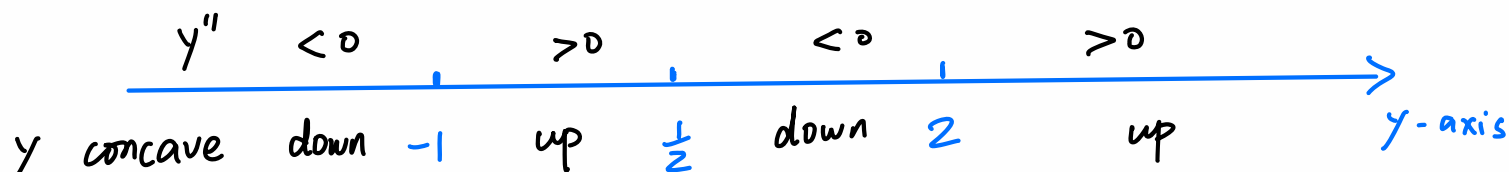
e.g. $\frac{dy}{dx} = (y+1)(y-2)$.

- roots of RHS = $\underbrace{-1}$ and $\underbrace{2}$.
equilibrium values



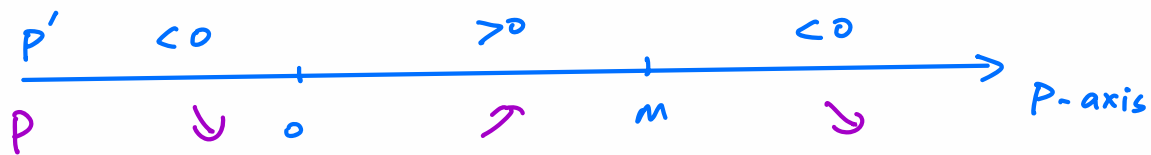
- One can further sketch the solutions by checking concavity:

$$y'' = [(y+1)(y-2)]' = (y+1)(y-2)(2y-1).$$

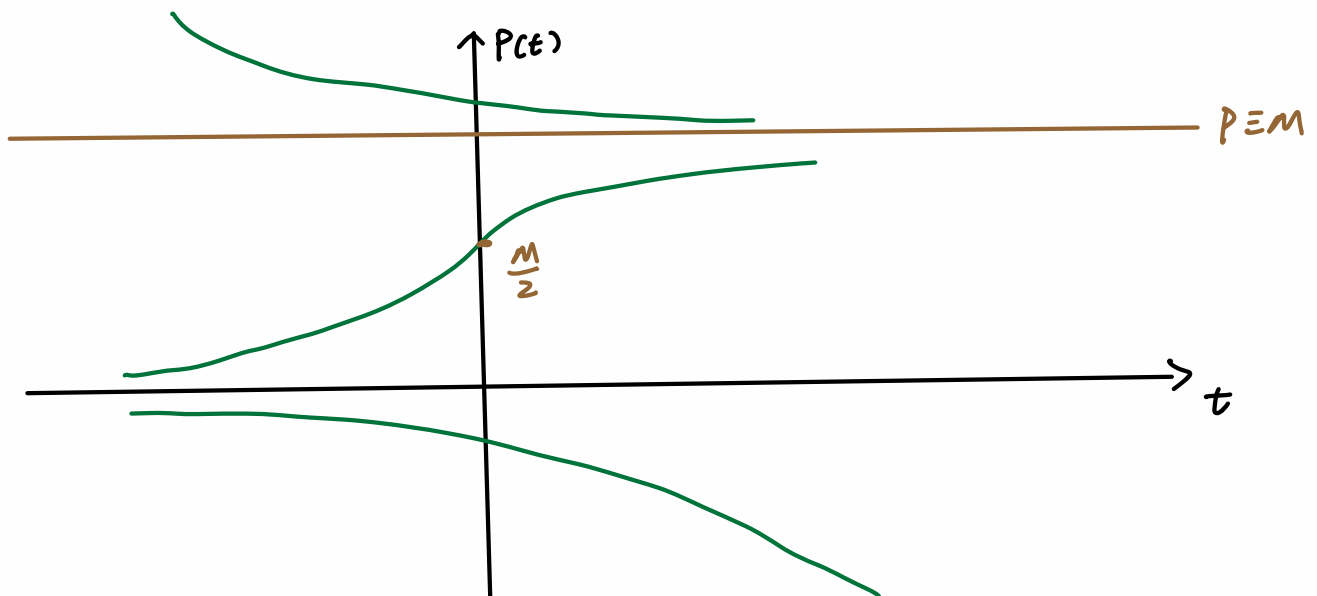
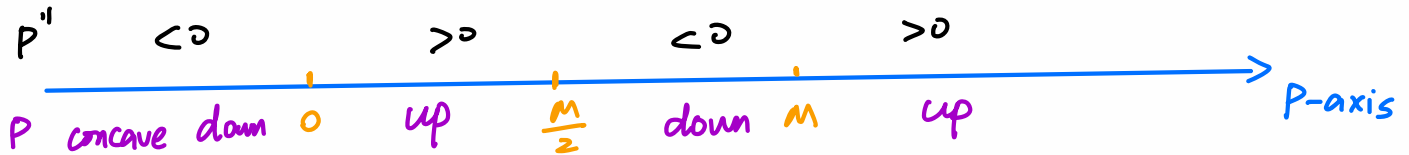


e.g. $\frac{dP}{dt} = kP \left(1 - \frac{P}{M}\right), \quad k > 0, M > 0.$

$kP \left(1 - \frac{P}{M}\right) = 0 \implies P = \underline{0} \text{ or } P = \underline{M}$
 equilibrium values



$\frac{d^2P}{dt^2} = k \left[P' \left(1 - \frac{P}{M}\right) + P \left(-\frac{1}{M}\right) P' \right]$
 $= kP' \left[1 - \frac{P}{M} - \frac{P}{M} \right]$
 $= k \left[kP \left(1 - \frac{P}{M}\right) \right] \cdot \left(1 - \frac{2P}{M}\right)$

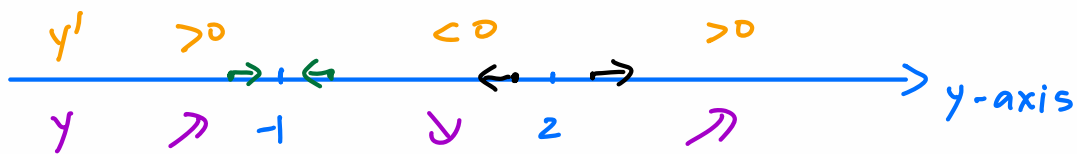


Practical meaning: at any given time:

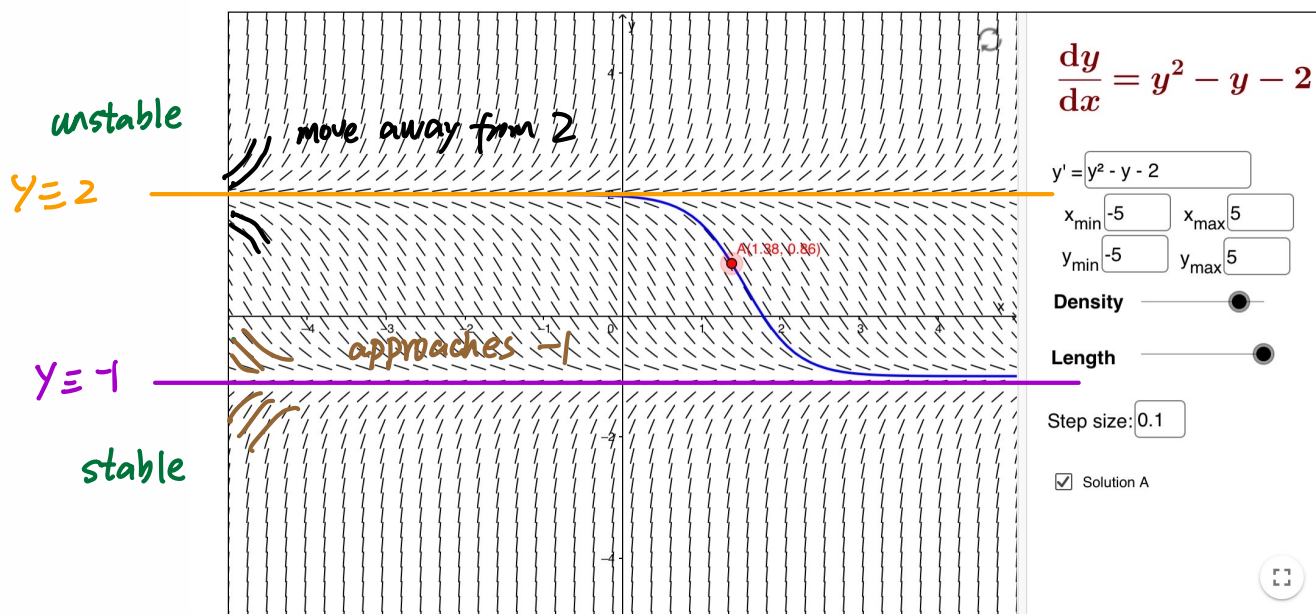
- if $0 < P < M$, then P will increase toward M ;
- if $P > M$, then P will decrease toward M ;
- if $P = 0$ or $P = M$, then P will stay unchanged. [equilibrium solutions].

Stable vs unstable equilibria

Consider the autonomous equation $y' = (y+1)(y-2)$.



- $y = -1$ is a **stable** equilibrium, since for any solution $y = g(x)$ with initial value $g(x_0)$ "near" -1 , $g(x) \rightarrow -1$ as $x \rightarrow \infty$.
- $y = 2$ is an **unstable** equilibrium, since for any solution $y = g(x)$ with initial value $g(x_0)$ "near" 2 , $g(x)$ moves away from 2 as x increases.



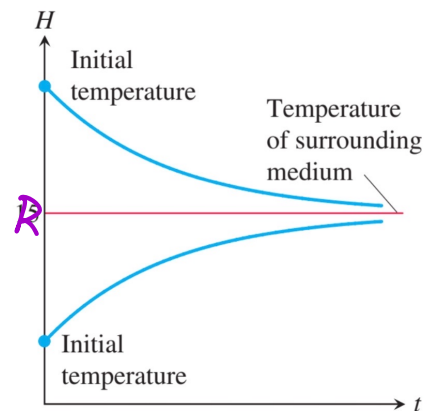
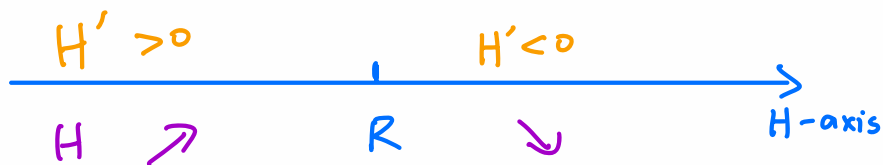
e.g. .

• For logistic D.E. $\frac{dP}{dt} = kP \left(1 - \frac{P}{M}\right)$,

$y \equiv M$ is stable while $y \equiv 0$ is unstable .

• For Newton's law of cooling $\frac{dH}{dt} = k(H - R)$ (where $k < 0$),

$H \equiv R$ is stable .



Euler's Method

• Suppose we want to solve an IVP $y' = F(x, y)$ with $y(x_0) = y_0$, but that we do not know the exact solution.

• Numerical methods help! One is Euler's method.

• The main idea:

Step 1: start with (x_0, y_0) .

Step 2: approximate the solution curve $y = y(x)$ by generating

a sequence of points $(x_0, y_0), (x_1, y_1), (x_2, y_2) \dots$ with tangent lines!

• Procedure:

Step 1: start with (x_0, y_0) .

— pick a small $\Delta x (> 0)$.

Step 2: set $\begin{cases} x_1 \triangleq x_0 + \Delta x \\ y_1 \triangleq y_0 + F(x_0, y_0) \cdot \Delta x \end{cases}$

$[\because y_1 \approx y_0 + f'(x_0)(x_1 - x_0)]$

Step 3: y_1 is meant to approximate

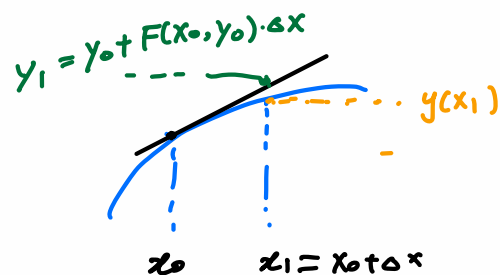
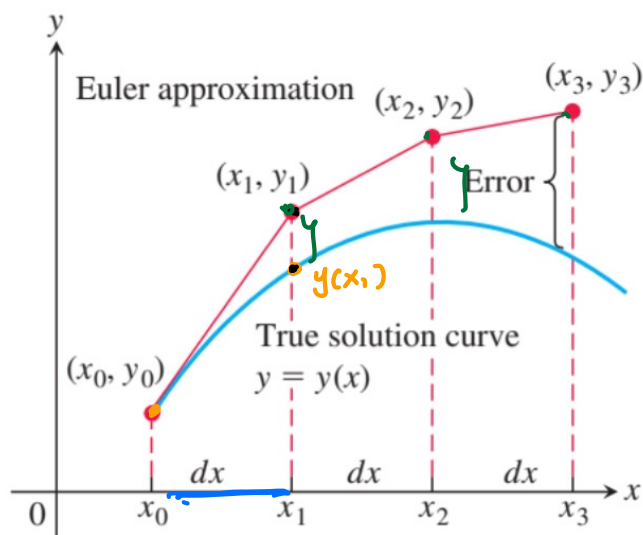
$y(x_0 + \Delta x)$, which is $y(x_1)$.

Step 4: At (x_1, y_1) , consider the line with

slope $F(x_1, y_1)$ (This is not a tangent line because

$y_1 \neq y(x_1)$ in general, but it is close to a tangent line when Δx is small).

Step 5: Set $x_2 \triangleq x_1 + \Delta x$, $y_2 \triangleq y_1 + F(x_1, y_1) \cdot \Delta x$.



Step 6: In general, for $k \geq 1$, set

$$\begin{cases} x_k \doteq x_{k-1} + \Delta x = x_0 + k \cdot \Delta x \\ y_k \doteq y_{k-1} + \Delta x \cdot F(x_{k-1}, y_{k-1}) \end{cases}$$

and y_k is taken as an approximation of $y(x_k)$.

e.g. Given $y' = 1 + y$ and $y(0) = 1$. Approximate $y(0.3)$ with $\Delta x = 0.1$.

Sol': $x_0 = 0$, $y_0 = 1$.

$$x_1 = 0.1, \quad y_1 = y_0 + \Delta x \cdot F(x_0, y_0) = 1 + 0.1 \cdot (1 + 1) = 1.2.$$

$$x_2 = 0.2, \quad y_2 = y_1 + \Delta x \cdot F(x_1, y_1) = 1.2 + 0.1(1 + 1.2) = 1.42.$$

$$x_3 = 0.3, \quad y_3 = y_2 + \Delta x \cdot F(x_2, y_2) = 1.42 + 0.1(1 + 1.42) = 1.662.$$

$$\therefore y_3 = 1.662.$$



Actual value : $y(0.3) = 1.6997 \dots$ if you solve the D.E. directly.

Exercise. Try to approximate a solution to

$$\begin{cases} y' = y \cdot \frac{\sin x}{x} \\ y(-0.5) = 2 \end{cases}$$

with Euler's method.

Part I. Limits and Continuity

1. Limits.

• Definition : ϵ - δ language

— existence ($\lim_{x \rightarrow a} f(x) = L$) : $\lim_{x \rightarrow a-} f(x) = L$ and $\lim_{x \rightarrow a+} f(x) = L$.

• Properties :

— uniqueness : if $\lim_{x \rightarrow a} f(x)$ exists ^{finitely}, then the value is unique (as L).

— "locally bounded" : if $\lim_{x \rightarrow a} f(x) = L$, then $L - \overset{0.1}{\delta} \leq f(x) \leq L + \overset{0.1}{\delta}$ locally.

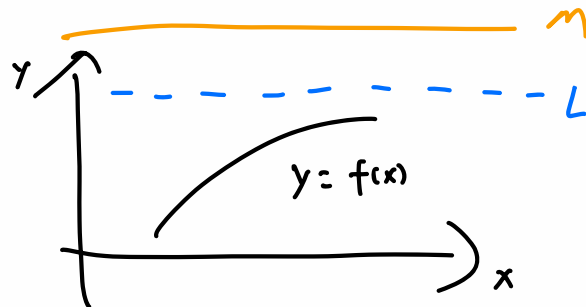
— "preserving the sign" : if $f(x) \geq 0$ and $\lim_{x \rightarrow a} f(x) = L$, then $L \geq 0$.

• Criteria of existence :

— Sandwich theorem

— Monotone bounded convergence theorem (optional) :

if $f(x)$ is increasing on (A, ∞) and $f(x) \leq M < \infty, \forall x$,
then $\lim_{x \rightarrow \infty} f(x)$ exists.



• Important cases :

— $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

— $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$

• Limit rules:

— $+, -, \times, \div$ (we require that in each step the limits exist!)

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \cdot h(x)$$

— composite functions: $\lim_{x \rightarrow a} f(g(x)) \stackrel{f \text{ cont}}{=} f(\lim_{x \rightarrow a} g(x)) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \lim_{x \rightarrow a} h(x)$

— L'Hospital rule $(\frac{0}{0}, \frac{\infty}{\infty})$ and $(\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ exists!})$

— equivalent replacing (product of quotients!) $\times, \div$

$$\left(\lim_{x \rightarrow 0} \frac{\sin x (1 - \cos x)}{e^x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1 - \cos x}{\frac{1}{2} x^2} \cdot \frac{x \cdot \frac{1}{2} x^2}{e^x} = 1 \cdot 1 \cdot \lim_{x \rightarrow 0} \frac{x \cdot \frac{1}{2} x^2}{e^x} \right)$$

— Big/Little Oh

(Relative rate of growth)

$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} \neq \lim_{x \rightarrow 0} \frac{x - x}{x^3}$$

2. Continuity

• Definition: $\lim_{x \rightarrow a} f(x) = f(a)$

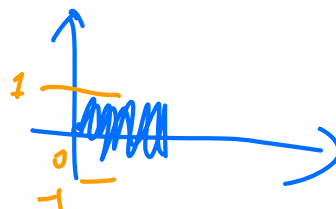
• Discontinuity

— removable, jump

— infinite: $f(x) = \begin{cases} \frac{1}{x}, & x \neq 0; \\ 0, & x = 0. \end{cases}$

— oscillating: $f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0; \\ 0, & x = 0. \end{cases}$

$$f(x) = \begin{cases} x \sin \frac{1}{x}, & x \neq 0; \\ 0, & x = 0. \end{cases}$$



• Asymptotes :



— horizontal : $\lim_{x \rightarrow \infty} f(x) = b$ or $\lim_{x \rightarrow -\infty} f(x) = b$

— vertical : $\lim_{x \rightarrow x_0^+} f(x) = \infty$ or $\lim_{x \rightarrow x_0^-} f(x) = \infty$ or ...

— oblique :

$$\left\{ \begin{array}{l} \lim_{x \rightarrow \infty} (f(x) - ax) = b \\ \lim_{x \rightarrow \infty} \frac{f(x)}{x} = a \end{array} \right. \quad \text{or} \quad \left\{ \begin{array}{l} \lim_{x \rightarrow -\infty} (f(x) - ax) = b \\ \lim_{x \rightarrow -\infty} \frac{f(x)}{x} = a \end{array} \right.$$

- Continuity with operations

- $+$, $-$, $\times$, $\div$

- inverse function

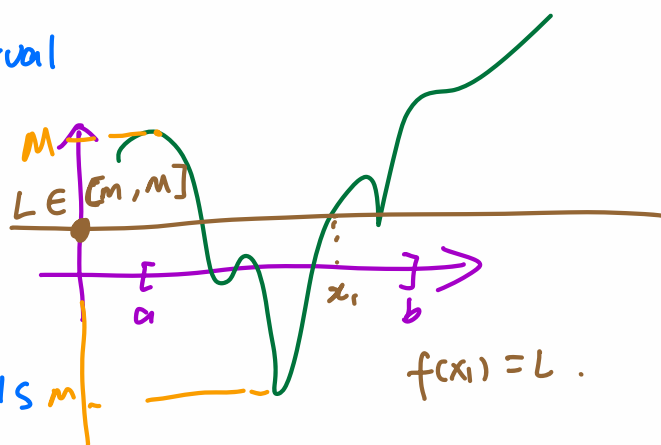
- composite functions

- elementary functions : $\ln$, $\sin$, $\exp$, $\tan$, ...

$\arctan x$

- Continuous functions on closed interval

- intermediate value thm



Part II. Derivatives and differentials

1. Derivatives

- Definition : $\lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$ [slope of the tangent line]

- differentiable $\Rightarrow$ continuous

- derivative formulas $(x^n)' = n x^{n-1}$, $(a^x)'$, $(\ln x)'$, $(\tan x)'$
 $(\arcsin x)'$, ...

- differentiation rules

- $+$, $-$, $\times$, $\div$

- inverse function: $(f^{-1})'(y) = \frac{1}{f'(x)} \Big|_{x=f^{-1}(y)}$

- composite function : chain rule
- implicit function : $f(x, y) = 0 \Rightarrow y' = ?$
- higher order derivatives $(x + y \sin x = 0 \Rightarrow 1 + y' \sin x + y \cdot \cos x = 0)$
- linearization : $f(x + \Delta x) \approx f(x) + f'(x) \cdot \Delta x$. (Δx very small!)

$$\begin{aligned} \Delta y &= f(x + \Delta x) - f(x), \\ dy &= f'(x) \cdot \Delta x = f'(x) dx \\ dx &= \Delta x \end{aligned}$$

- Mean value thm



$$\text{— Rolle's thm : } \left. \begin{array}{l} f \text{ cont } [a, b] \\ f \text{ diff } (a, b) \\ f(b) = f(a) \end{array} \right\} \Rightarrow \exists c, f'(c) = 0 \quad \Rightarrow$$

$$\text{— (Lagrange's) Mean value thm : } \exists c, \text{ s.t. } \frac{f(b) - f(a)}{b - a} = f'(c).$$

2. Applications

- monotonicity

- extreme value analysis

— local vs absolute



— critical points : $f'(x_0) = 0$ or $f'(x_0)$ does not exist.

— endpoints

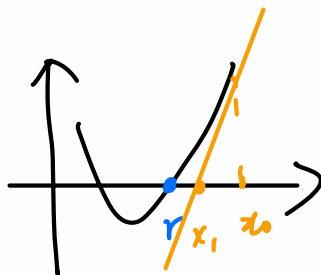
- Concavity

- analysis via $f''(\cdot)$

- inflection point.

- Applied optimization

- Newton's method:



$$y = f(x_0) + f'(x_0)(x - x_0)$$

$$\stackrel{y=0}{\Rightarrow} x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Part III. Integration

1. Indefinite integral

- Antiderivative: $\int \underline{f(x)} dx = F(x) + C$. [$\underline{F'(x)} = f(x)$]

- Properties

- linearity: $\int \alpha f(x) + \beta g(x) dx = \alpha \int f(x) dx + \beta \int g(x) dx$

- elementary forms: $\int \tan x dx$, $\int \sec x dx$, $\int \ln x dx$, --

- Computation

- Substitution: $\int f(g(x)) g'(x) dx = \int f(u) du = F(u) = F(g(x))$

- Integration by parts: $\int f(x) g'(x) dx = f(x) g(x) - \int f'(x) g(x) dx$
[decreasing the degree!]

- rational functions: $\int \frac{P(x)}{Q(x)} dx$

$$\textcircled{1} \int \frac{A}{x-a} dx \quad \textcircled{2} \int \frac{A}{(x-a)^n} dx \quad \textcircled{3} \int \frac{A}{(x^2+a^2)^n} dx \quad \textcircled{4} \int \frac{Ax+B}{(x^2+a^2)} dx = \int \frac{C}{x^2+a^2} dx$$

$$\textcircled{5} \int \frac{x+A}{(x^2+a^2)^n} dx = \int \frac{x}{(x^2+a^2)^n} dx + \int \frac{A}{(x^2+a^2)^n} dx$$

- trigonometric functions: $\int \sin^m x \cos^n x dx$, $\int \tan^m x dx$,

- trigonometric substitution: $\int \tan^m x \sec^n x dx$, $\int \frac{1}{\sqrt{1+4x^2}} dx$

$$\int \underbrace{\sqrt{a^2 - x^2}}_0 dx, \int \underbrace{\sqrt{a^2 + x^2}}_0 dx, \int \sqrt{x^2 - a^2} dx$$

2. Definite integral

• Definition

- "Area of a shape"
- limit of all Riemann sums
 - partition, left/mid/right-point rule, ...
- Integrable: bounded? continuous? continuous with finite jumps?

• Properties

- Linearity ✓
- Additivity: $\int_a^c f(x) dx =$
- "Preserving the sign": $f(x) \leq g(x), \forall x \Rightarrow$
- integration mean value thm: $\exists C$, s.t. $\int_a^b f(x) dx =$
- FTC 2: $\int_a^b f(x) dx = F(x) \Big|_a^b$.
- FTC 1: $(\int_a^x f(t) dt)' = f(x)$.

• Improper integral: $\int_a^\infty f(x) dx, \int_a^b f(x) dx \quad [\lim_{x \rightarrow a^+} f(x) = \infty]$

— Convergence test: Direct, Limit

Compare size: $\int_a^\infty f(x) dx$ vs $\int_a^\infty g(x) dx$.

3. Applications

- Area : $\int_a^b \underbrace{|f_1(x) - f_2(x)|}_{\Delta} dx$

- Volume :

- Cross-sections :

$$V = \int_a^b A(x) dx .$$

- cylindrical shells :

$$V = \int_a^b 2\pi x \cdot h(x) dx$$

- revolution :

$$V = \int_a^b \pi [R(x)^2 - r(x)^2] dx .$$

- Arc length

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx .$$

- Area of surfaces of revolution

$$S = \int_a^b 2\pi f(x) \cdot \sqrt{1 + [f'(x)]^2} dx .$$

- Work and Fluid forces

$$W = \int_a^b F(x) dx , \quad F = \int_a^b \underbrace{w \cdot (\text{strip depth})}_h \cdot \underbrace{L(y) dy}_A$$

- Definitions of transcendental functions

- $\ln(x), \exp(x), a^x, \log_a x$.

$$\int_1^x \frac{1}{t} dt$$

- numerical integration

- Left / Mid / Right-point rule

- Trapezoidal rule

$$\sum_{k=1}^n \frac{f(x_{k-1}) + f(x_k)}{2} \Delta x = \frac{h}{2} (y_0 + 2y_1 + \dots + 2y_{n-1} + y_n)$$

- Simpson's rule

$$\frac{h}{3} (y_0 + 4y_1 + 2y_2 + \dots + 2y_{n-2} + 4y_{n-1} + y_n),$$

Part IV. Differential equation

- Definitions : $G(x, y, y') = 0$.
 - solution $y = y(x)$
 - slope fields
 - Method of separation: $\frac{dy}{dx} = f(x)g(y)$.
- First-order linear D.E. $\frac{dy}{dx} + P(x) \cdot y = Q(x)$
 - Method of integrating factor: $e^{\int P(x) dx} (y' + P(x)y)$
- Applications : Malthus, Logistic, Newton's law of cooling
- Autonomous equations: $\frac{dy}{dx} = f(y)$.
 - phase line analysis
 - equilibrium solution
 - Stable solutions
- Euler's method :
$$\begin{cases} x_k \triangleq x_0 + \Delta x \cdot k, \\ y_k \triangleq y_{k-1} + \Delta x \cdot F(x_{k-1}, y_{k-1}), \end{cases}$$